

Prediction Algorithm Accuracy Report — v4

Season-Boundary Elo Reset

Sports Analytics Project

September 2026

Abstract

The v3 report flagged an explicit, deliberately-unaddressed simplification in the Elo model: `compute_elo_ratings` treated every game across all three loaded seasons as one continuous chronological sequence, with no reset or regression-toward-the-mean at season boundaries, even though real NBA team strength shifts over an offseason (trades, draft, free agency). This report backtests and ships a fix. Holding the already-validated `HOME_COURT_ADVANTAGE_ELO` = 40 and `K_FACTOR` = 25 constants fixed, a season-boundary reset that regresses each team's rating 50% of the way back toward the 1500 starting point was tested in isolation and found to be a real, robust improvement: it wins on two independent chronological validation splits, on 19 of 20 randomized split points spanning a wide range of train/validation cut points, and against a permutation-test null ($p < 0.001$). `SEASON_RESET_FRACTION` = 0.5 is now shipped.

Contents

1	Motivation	1
2	Method	2
2.1	Isolating the reset from the already-shipped tuning	2
2.2	Grid search and validation	2
2.3	Characterizing the shape of the effect before picking a shipped value	2
2.4	Additional robustness checks	3
3	What Shipped	3
4	What Was Deliberately Not Done	4
5	Recommendations Going Forward	4

1 Motivation

v3's Elo re-tuning (`HOME_COURT_ADVANTAGE_ELO`, `K_FACTOR`) was deliberately run against the existing no-reset model so that experiment isolated one variable — more data — rather than two. That report's module-docstring caveat named the season-boundary question explicitly as the natural next, separately-evaluated step. This report is that step.

2 Method

2.1 Isolating the reset from the already-shipped tuning

HOME_COURT_ADVANTAGE_ELO and K_FACTOR were held fixed at their v3-validated values (40 and 25) throughout this experiment. Only a new parameter, SEASON_RESET_FRACTION, was grid-searched — the fraction of the distance back toward STARTING_ELO (1500) that every team’s rating regresses whenever a game’s season differs from the previous game’s. SEASON_RESET_FRACTION = 0.0 exactly reproduces the pre-this-change behavior (verified programmatically: a 0.0-fraction pass was diffed against elo.py’s actual compute_elo_ratings output game-by-game across all 3,781 games before any grid-search result was trusted).

2.2 Grid search and validation

The same chronological train/validation discipline as v3’s HCA/K tuning was applied: a grid over SEASON_RESET_FRACTION $\in [0, 0.5]$ was searched on each split’s training portion (by Brier score), and the winner was checked against that split’s held-out validation portion.

Split	No reset (0.0) validation Brier	Best-on-train reset validation Brier
70/30 chronological	0.2119	0.2095 (fraction 0.40)
60/40 chronological	0.2101	0.2077 (fraction 0.45)

Table 1: Reset beats no-reset on both independent validation splits.

2.3 Characterizing the shape of the effect before picking a shipped value

The two splits’ individual best-on-train fractions (0.40 and 0.45) did not exactly agree, and the training-split Brier surface near those points was quite flat (0.2156–0.2165 across fractions 0.30–0.50 on both splits) — the same pattern v1’s single-season HCA/K attempt showed before turning out to be noise. Given that history, the fraction wasn’t trusted at face value; the full validation-Brier curve was examined instead of just each split’s single nominal winner.

Fraction	70/30 split: validation Brier	60/40 split: validation Brier
0.00 (no reset)	0.2119	0.2101
0.20	0.2104	0.2086
0.40	0.2095	0.2078
0.50	0.2092	0.2076
0.60	0.2090	0.2077
0.70	0.2089	0.2079
0.80	0.2090	0.2083
1.00 (full reset)	0.2093	0.2096

Table 2: Validation Brier across the full fraction range, both splits. Both curves fall monotonically from 0.0 to roughly the 0.5–0.7 region, then rise again toward a full reset at 1.0 — a genuine interior optimum, not noise or a boundary effect. The two splits’ minima (0.70 and 0.50–0.60 respectively) are close but not identical.

This is materially different from — and more convincing than — a single sharp per-split “winner”: both curves are monotonically decreasing from no-reset, bottom out in a shared broad region, and

rise again toward a full reset, on both splits independently. **Some reset clearly beats none, and a full reset is clearly worse than a partial one, on both splits**, even though the single sharpest point differs slightly between them.

`SEASON_RESET_FRACTION = 0.5` was chosen as the value inside both splits’ shared near-optimal region, rather than either split’s individual point-estimate — deliberately not overfitting to one split’s specific nominal winner (0.40 or 0.45) when the two disagreed by that much and the surface was this flat around the optimum.

2.4 Additional robustness checks

Two checks beyond the two-split comparison were run before shipping, matching the standard this project has applied to every prior tuning decision:

- **Many randomized split points.** Rather than trusting two fixed splits (70/30, 60/40), 20 additional chronological train/validation splits were generated with the split point drawn uniformly at random from the 50%–85% range of the season timeline. `SEASON_RESET_FRACTION = 0.5` beat no reset on **19 of 20** of these splits.
- **Permutation test.** 1,000 reruns of the full 3,781-game schedule with each game’s outcome replaced by an independent coin flip (same methodology as every permutation test in this report series) produced a null Brier distribution with mean 0.2609 and standard deviation 0.0013; the observed Brier score at `SEASON_RESET_FRACTION = 0.5` (0.2130) beat every single one of the 1,000 null runs ($p < 0.001$).

3 What Shipped

`elo.py`’s `compute_elo_ratings` now detects a season boundary (the current game’s `season` differing from the previous game’s) and regresses every team’s current rating 50% of the way back toward 1500 before processing that boundary game and the rest of its season. `predict_upcoming_games` applies the same reset when every completed game so far belongs to an earlier season than the upcoming games being predicted — otherwise the very first predictions of a brand-new season (before any of its games have been played) would use stale, un-reset ratings until the first real result triggered the reset organically.

`GamePrediction` was regenerated for all 3,781 games under the new model.

Metric	Before reset (v3)	After reset (v4)
Full-dataset Brier score	0.2145	0.2130
Full-dataset accuracy	66.0%	65.8%

Table 3: Full-dataset (non-held-out) figures, before vs. after. As with the v3 HCA/K tuning, the headline full-dataset number moves modestly — the real evidence for the change is the held-out validation-split comparison in the tables above, not this summary statistic.

Six unit tests (`test_elo.py`) cover the reset’s exact semantics: it fires once and only once per season boundary (not repeatedly within a season), it’s a true no-op for a team with no rating history yet, it regresses by exactly the configured fraction, and `predict_upcoming_games` correctly applies a pending reset for a season whose first game hasn’t been played yet. All pass, alongside the full existing suite (187 tests across the predictor, API, and web apps) with no regressions.

4 What Was Deliberately Not Done

The reset baseline is a fixed global constant (1500), not each team's own multi-season average rating or any per-team prior — matching the recommendation from v3 to keep this experiment testing one mechanism at a time. A team-specific baseline (e.g. a team's 3-year rolling average rather than the league-wide 1500 starting point) is a reasonable further refinement but was out of scope here, since it would reintroduce exactly the kind of multi-variable conflation this report was structured to avoid.

5 Recommendations Going Forward

1. **This closes the caveat v3 opened.** Elo's model structure (constants, season-awareness) is now validated as thoroughly as the available data supports; further Elo gains most plausibly come from more data (more seasons) rather than more model refinement, echoing v3's own recommendation.
2. **Four Factors and the player-point predictors remain the least-improved models relative to their naive baselines** across this entire report series. None of the season-boundary logic here was applied to Four Factors' running-average computation, which has the same implicit continuous-sequence assumption Elo had — worth a similarly isolated, backtested experiment as a future step, now that the methodology (isolate one variable, validate on multiple splits plus randomized splits plus a permutation test) has been exercised three times running.

Appendix: Reproducing These Results

`apps/predictor/check_accuracy.py` recomputes Elo's Brier score/accuracy against the database's current state, reflecting the shipped `SEASON_RESET_FRACTION` automatically since it calls the real `compute_elo_ratings` directly. The grid-search, randomized-split, and permutation-test scripts behind this report are one-off analysis, not part of the committed codebase, matching the same appendix note in v3.